

# ● House Price Prediction: Analysis and Business Insights

- This project aimed to predict house prices using property features such as area, bedrooms, bathrooms, stories, parking, furnishing status, and various amenities. Exploratory Data Analysis revealed that **area is the most influential factor affecting house prices**, followed by bathrooms, stories, parking availability, and preferred location. Houses with air conditioning, guest rooms, basements, and better furnishing status generally had higher average prices. The analysis also showed the presence of a luxury housing segment, represented by a few high-priced outlier properties.
- Two machine learning models were trained and evaluated: **Linear Regression** and **Random Forest Regressor**. Linear Regression achieved an MAE of approximately ₹970,043 and an  $R^2$  score of 0.653, while Random Forest achieved an MAE of approximately ₹1,021,151 and an  $R^2$  score of 0.612. Since Linear Regression produced lower prediction error and explained a larger portion of price variation, it was selected as the better-performing model for this dataset. Additionally, its simplicity and interpretability make it more suitable for business applications.
- Several interesting insights emerged from the analysis. Contrary to common belief, the number of bedrooms alone was not the strongest determinant of price; house area and location had a much greater impact. Properties located in preferred areas consistently achieved higher prices, highlighting the importance of location in real estate valuation. Premium amenities such as air conditioning, parking facilities, and furnished interiors also contributed significantly to property value.
- Based on these findings, real estate developers seeking to maximize house prices should focus on constructing larger homes in premium locations while incorporating high-value amenities such as multiple bathrooms, parking spaces, air conditioning, guest rooms, and modern furnishing. Targeting the luxury housing market with spacious, well-equipped properties can further increase profitability. Overall, the project demonstrates how data-driven insights and machine learning can support better pricing strategies and investment decisions in the real estate industry.